1.1 PROJECT DESCRIPTION

Face recognition has become a key research area in the rapidly growing world of science and technology. Traditionally identifying repeat offenders relied on biometric methods like fingerprint or handprint analysis. However criminal tactics have evolved many offenders are now careful not to leave physical evidence at crime scenes. This change limits how effective these older methods can be. To tackle this challenge many well developed countries have earlier created and maintained large facial image databases. These databases let authorities match unknown or suspicious individuals against stored records speeding up the identification process and making it more reliable.

In our system we begin by taking a face image either captured from CCTV footage during a crime or a photo provided for a missing child. The system then automatically analyses the image and searches for a match in the existing database. If it finds a match it displays the person's details right away.

This approach helps law enforcement quickly identify and track suspects. It also acts as a bridge of hope that brings lost children back into the embrace of their families. By combining smart technologies that improves processing the system works faster and more accurately. This makes it a trusted shield that quietly safeguards todays world and safety efforts.

❖ Problem Statement

At its heart this project aims to build a smart face recognition system which people use afar making it both practical and easy to reach registration, monitoring and identification of criminals and missing children using digital image data. Today law enforcement and child protection agencies face increasing pressure to identify individuals quickly often in real-time. Traditional manual methods can be slow and not work good specially in busy situations where every second counts. The project shows that challenging by providing a modern, automated and reliable solutions using Computer visions and deep learning.

The system takes an image as input. This could be a photo of a suspect captured on CCTV during a crime or a picture of a missing child shared by family members or child protection authorities. First the image is pre-processed for better clarity and converted to grayscale. using OpenCV the system detects and isolates the face removing unnecessary background details to focus entirely on the facial region. Once the face is isolated the system extracts features. The Local Binary Pattern method captures subtle details of facial textures which are unique to each person. These details serve as a digital signature for the individuals face.

Next A convolutional networks process features and compares them with the database of known criminals and registered missing children. Finally the system provides a result indicating whether a match is found. If a match exists the system instantly retrieves the persons stored details such as identity or case information. This allows law enforcement to quickly track suspects involved in criminal activities and helps locate and help lost child find way into the arms of their loved ones. By merging tradition image process technique with deep learning models and this project offers a fast, accurate and scalable solutions which meets today's need for security law enforcement and child safety.

❖ Objective of the Study

- What this project is really set out to do is to support law enforcement agencies and child protection services by shaping a smart way of recognizing faces that can identify both criminals and missing children. Traditional identification methods like manual verification or fingerprint analysis, are often slow, unreliable and less effective especially when criminals change their appearance to avoid being caught. Our way is to tackles challenge by leaning on deep learning where image-based models guide the solution ensure quicker and more accurate identification.
- This has been works together to quickly analyse and identify faces by comparing an input image with a database of known individuals. Unlike older methods this application can still recognize faces despite small changes in appearance such as different hairstyles, facial hair or lighting conditions. This design makes the system useful for real-world situations where captured pictures may not always be sharp yet they play role in the helping trace lost kids as it can match their facial features with stored records even after minor changes in age or looks.
- By automating the face recognition process the system greatly reduces the need for manual effort and reliance on human input in investigations. This automation speeds up the workflow allowing quicker decision-making for police and security forces. The time saved can be vital in criminal investigations and in bringing lost kids back to the warmth of their family circle. Overall the application improves the efficiency, accuracy and reliability of law enforcement operations while providing a modern solution to an important social issue.

❖ **Scope of the Project**

Face recognition across different ages, appearances and environmental conditions is counted among the toughest tasks in computer vision. A number of thinkers have put forward different ideas ways to improve accuracy. However a noticeable gap remains in achieving consistent high performance when faces change due to aging, hairstyle variations facial hair growth or differences in lighting and pose. The success of a face recognition system relies heavily on the choice of Feature Extraction and classification methods.

The proposed system tackles these challenges by using advanced learning approaches especially those that mimic how the human brain reads images like CNNs. These networks first pick out the important details and then decide what they belong to in one framework. This allows the system to automatically learn and recognize complex facial patterns without needing manual feature development. As result the system becomes stronger and easier to real world conditions.

The scope what this project aims for is more than simply research. It has practical uses in several areas:

- Law Enforcement: Helps police departments identify criminal suspects from CCTV footage and aids in tracking and bringing lost kids back into the care of their families.
- Education Sector: finds its place in automated attendance systems in schools, colleges and universities saving time and preventing proxy attendance.
- Corporate & Workplaces: Improves security and efficiency in organizations by enabling employee identification and attendance monitoring.
- Banking & Hospitality: Boosts security in banks, hotels and public service areas by ensuring accurate identification and reducing fraud.
- Airports & Transportation Hubs: Assists with passenger verification and monitoring leading to safer and smoother travel.

Overall, the proposed system has a wide reach and can greatly enhance security, monitoring and identification processes in both public safety and commercial areas. Its ability to scale and adapt makes it solution for real world challenge.

2 LITERATURE SURVEY

2.1 Existing and Proposed System

❖ **Existing System**

Face recognition has come a long way from where it started a lot and is now widely used to identify criminals and find missing children. Various existing systems use face recognition methods primarily Brain inspired models built on convolutional networks to achieve high accuracy and help lower crime rates. These systems create unique facial templates and compare them to large databases providing relevant details when a match is found. Using CNN in face recognition enables fast and precise detection with minimal computing time making real-time applications possible in security and law enforcement.

Systems are designed to work effectively with both images and videos with some implementations achieving accuracy levels of over 90%. In certain case real-time camera feeds such as CCTV and mobile cameras are integrated improving the system's flexibility and responsiveness. Datasets like FGNET, MORPH-II and Flickr-Faces-HQ have been helping check how well something actually works CNN based models. These models do not need complex preprocessing steps like correcting head pose which streamlines the process. Lightweight models using small image sizes are known to be both resource efficient and accurate.

The criminal identification process benefits from key image processing methods such as contrast improvement, thresholding and feature scaling. Some systems update dynamically with new data from online sources allowing for continuous improvement. Face recognition is seen as a steadier option compared to traditional biometric methods such as thumbprints. This is especially true as modern criminals become more cautious about leaving physical evidence.

To boost recognition performance, hybrid approaches that combine CNN with People have also recommended type of classifier support vector machines resulting in better outcomes than standard neural networks. Overall these systems support quicker identification, reduce crime and make communities safer without getting in the way memory efficient and suitable for large scale use.

❖ **Limitations:**

- Vulnerability to Spoofing: Although the way computers can recognize faces has gotten a lot better significantly a big hurdle to tackle is their vulnerability to spoofing attacks. Criminals can use images or videos from social media to trick recognition systems if proper liveness detection or anti spoofing measures are not in place. Without these protections the system may incorrectly identify or authenticate individuals leading to serious security risks.
- Environmental and Facial Variations: How spot on something is face recognition declines noticeably in poor lighting when someone's face isn't fully visible or when significant changes occur due to aging, different poses or disguises. Many models tend to be overfitted to specific datasets and may not perform well in diverse real world situations. For example blurry CCTV footage or poor or low quality image are resulted in performance levels that do not meet practical or acceptable standards.
- Ethical and Privacy Concerns: The use of face recognition systems raises important ethical and legal issues especially around privacy, data security and the misuse of biometric information. Unauthorized surveillance, data leaks or improper use of facial data can violate personal rights and lead to public distrust in these technologies. Many regions are still developing proper guidelines, regulations and policies.
- High Resource Requirements: While CNN-based these smart learning systems reach impressive levels of accuracy they need big collections of labelled images ready to teach the system and powerful computing Resources things like GPUs and TPUs. This makes the building, shaping and implementation resource intensive and expensive particularly for teams working with tight budgets.
- Scalability Issues: Current systems often rely on limited local databases. Without proper integration into national or global databases their scalability and usefulness for large scale law enforcement or international criminal tracking remain limited. A broader data sharing framework is key for the system to reach its full potential.

❖ **Proposed system**

The system we are putting forward for tracking criminals and missing children uses using smart learning methods to make it faster and more precise face recognition. The system is powered by CNN a type of smart AI that understands images which excel at extracting detailed facial features from images. CNNs can detect patterns in pixel data that humans struggle to notice which makes them perfect for spotting people. The system uses this capability to enhance identification for both criminals and missing children. CNNs substantially reduce recognition errors by learning complex features like the shape of the face, eyes, nose and mouth which remain consistent even under various lighting conditions and angles.

To further boost model performance and efficiency, the proposed system uses pretrained transfer learning. Pretrained models like VGGFace, FaceNet and ResNet which have learn by large and diverse facial datasets are fine-tuned for the specific needs of the identification system. By using transfer learning, the system cuts down on the time and data required for training. Instead of starting from scratch it builds on the knowledge gained from these established models. This speeds up the training process and ensures the system benefits from extensive and diverse data these models have encountered resulting in more accurate and reliable outcomes.

A significant challenge in recognizing missing children is facial aging, as people faces change over time. To tackle proposed system includes many different deep learning based age progression techniques. These techniques simulate how a childs face might evolve as they grow up. By understanding how a persons face shifts or changes over age the system can accurately match older photos to current images even after several years. This feature is crucial situations in which kids ended up missing for extended periods as it improves the chances of identification by providing age invariant recognition.

Additionally the system boosts its learning by creatively expanding the data it seems to boost its robustness and adaptability. Real world images often vary due to differences in lighting, angles, resolution and other factors. The system is able to handle such variations ways to mix up the data such as turning images around flipping and adjusting brightness are used on artificially enlarge the training dataset. These techniques help the model become more resilient to common image distortions ensuring it can still make accurate identifications even when faced using pictures that are not the same as the others used for training.

❖ **Advantages:**

- Improved accuracy in identification: The system makes use of CNNs a type of AI that understands images which pulls out detailed face traits to use for highly accurate recognition. This results in fewer errors in identifying criminals or missing children even when the surroundings keep changing like lighting and angles.
- Enhanced recognition of aged faces: With deep learning-based age progression techniques, the system can identify missing children even after years by simulating how facial features change. This significantly boosts the chances of successful identification over time.
- Reduced training time with transfer learning: Using pretrained models like VGGFace, FaceNet, and ResNet allows the system to benefit from existing knowledge. This reduces the time and data needed for training enabling faster deployment all the while keeping top-notch accuracy.
- Real time identification: The system is to built work instantly face recognition allowing law enforcement to quickly track criminals and find missing persons. This reduces delays and improves efficiency in urgent situations.
- Reduced manual monitoring: Automating face recognition cuts down the need for manual surveillance which is time consuming and prone to errors. This saves time and reduces human mistakes, allowing law enforcement to concentrate on crucial tasks.
- Higher efficiency in crime prevention: The real time face recognition and quick processing enable rapid responses to criminal activities. This leads to faster identification and arrests enhancing crime prevention and public safety.

2.2 Feasibility Study

❖ **Technical Feasibility:**

Technical feasibility refers to evaluating whether the Face Recognition System for Criminals and Missing Children can be successfully developed with the available technologies, tools and resources. In this project it means checking if the system requirements such as hardware and database integration can be effectively used to build the application.

The proposed system uses Python, OpenCV, TensorFlow, Keras and MTCNN. These are reliable technologies for face detection and recognition. The system relies on CNN based models that are effective at learning distinct facial features. By using open source and affordable tools it avoids costly licensing fees and benefits from large developer communities that provide regular updates and improvements. For performance the system runs efficiently on mid-range GPU systems during training and recognition tasks. It can also scale through cloud platforms like AWS and Google Cloud for real time processing. Integration is possible through APIs and databases such as MySQL or Firebase. This system allows easy integration with criminal records, missing child databases and law enforcement systems. It supports deployment across various platforms including desktop web and mobile applications and works with inputs from CCTV cameras, webcams and mobile devices.

The design is scalable. It can handle growing amounts of data increased user traffic and future upgrades without compromising performance. The project is technically feasible because all the necessary technologies such as Python, CNNs, OpenCV, TensorFlow and cloud services are readily available, reliable and cost effective. The system can be implemented using existing tools integrated with current databases and scaled for real world applications. Since it does not rely on any rare or specialized technology it can be successfully developed and deployed with the resources now available.

❖ **Economic Feasibility:**

Economic feasibility means assessing whether a project can make money and is worth the costs. It looks at total development costs, operational expenses, maintenance needs and the potential benefits or profits the system might bring. When it comes to spotting faces for Criminals and Missing Children economic feasibility ensures the system is able to set up without high costs while still providing important value to law enforcement and child protection agencies.

The project is financially sensible because it uses free, open-source tools like Python, OpenCV, TensorFlow and Keras which remove licensing fees. The system can run midrange GPUs or low cost cloud services, keeping hardware costs low. The application is also lightweight. It only needs a small development team of

two to three people and requires little maintenance which makes it cost effective over time. By helping law enforcement quickly identify criminals and find missing children the system cuts operational costs, shortens investigation times and provides a solid return on investment.

In summary the project is a practical financial choice. It pairs affordable tools and hardware with a small team and offers significant long term advantages. Its ability to speed up identification and boost efficiency makes it a valuable solution for law enforcement agencies.

❖ **Operational Feasibility:**

Operational feasibility looks at whether the system can help with effectively by its intended people without getting in their way current workflows. It makes sure the application is practical easy to use and efficient while meeting organizational needs. When the face recognition system for criminals and missing children operational feasibility confirms that police and administrators can easily use the system get accurate and timely results and integrate smoothly with existing surveillance tools and databases. This helps improve efficiency and supports effective investigations.

The Face Recognition System for Criminals and Missing Children has user friendly interface. This allows police officers and administrators to use it with little technical training. It uses automation and AI based accuracy to lessen manual work leading to quicker and more accurate identification. The system works in real time, processing webcam feeds or uploaded images immediately for urgent responses. Its expandable database allows for regular updates of records keeping information current and reliable. It integrates well with CCTV surveillance systems enabling continuous scanning of public areas. Additionally, the system can operate in multiple modes. It works online for full features and offline for basic matching, ensuring continuous usability in various field conditions.

The system is easy to use works well with current infrastructure and provides real-time results. Its automation cuts down on manual work and speeds up investigations. Features like database expansion and multiple mode operation ensure it remains useful. Overall system serves law enforcement and child protection agencies making it effective and reliable in real world operations.

2.3 Tools and Technologies Used

❖ **Python**:

Python is a high level interpreted programming languages which is popular in Artificial Intelligence, Machine Learning and computer vision. It is known for being simple, readable and well supported by a strong community. python has large set of libraries frameworks for tasks like image processing machine learning deep learning and graphical user interface development.

Python is programming language for the face recognition system. it processes image input preprocesses data detects faces extracts feature and classifies images. its compatibility with opencv allows efficient image processing it easily integrate deep learning models like VGGface for accurate face recognition. python also supports gui components like Tkinter for creating user friendly interfaces. it provides tools for file handling database interactions making it strong and flexible base for developing the entire system.

❖ **Open CV**

Open CV or Open source Computer Vision Library, is a popular open source toolkit for real time image and video processing. Open cv offers variety of tools and algorithms tasks face detection object recognition motion tracking image filtering many computer vision applications. It works with several programming languages such as python c++ and java. its efficiency strong performance and active community support make it popular in research and industry.

In the face recognition system opencv is crucial in the early stages of image processing. it converts images to grayscale detects and isolates facial areas and prepares the images before sending them to deep learning model for feature extraction and classifications. with support for haar cascades and other detection algorithms opencv accurately locates and extracts faces even in complex or cluttered images. It also allows for efficient reading manipulation visualization of images. This capability simplifies the handling of large datasets from cctv footage or uploaded photos. By preparing high quality inputs for deep learning model, opencv improves the accuracy of the system in identifying criminals and finding missing children. its efficiency make it part of the overall face recognition process.

❖ **MTCNN**

MTCNN or Multi-task Cascaded Convolutional Neural Network is a deep learning based face detection framework designed to detect faces and locate facial landmarks in images. Unlike traditional face detection methods MTCNN is highly effective across different facial poses, expressions and lighting conditions. It employs a three stage cascaded network architecture to refine face outlined areas and landmark positions

including the key face feature eyes, nose, mouth and chin. This multi task approach allows MTCNN to detect faces and localize landmarks simultaneously providing precise information about the location and the way it is positioned faces in complex images. What it can do manage occlusions and rotations and varying scales make it dependable option to real world face recognition application.

In the face recognition system MTCNN accurates detects crops face region from input image do not matter where they from cctv footage webcams and uploaded photographs. By precisely identifying facial landmarks MTCNN ensures that extracted face is aligned and normalized which greatly help to get precise result and for recognition model which follows. This stands in particularly crucial in real world situation where faces may be partially covered and rotated and aged. Accurate face detection by mtcnn reduces errors in recognition process and supports consistent identification of criminal and missing child. Its strong performance in various conditions make it key piece of system providing solid basis for effective face recognition.

- **VGGFace (ResNet50)**

VGGFace is a deep convolutional networks model created for face recognition tasks. It has learned from a big collection of dataset facial images. This training allows it to learn and extract deep facial embeddings that uniquely represent an individuals facial features. The ResNet50 based version of VGGFace improves performance by using residual connections. These connections help the network learn more complex patterns and avoid problems like vanishing gradients. This model can capture subtle variations in facial features. It remains effective against changes in expressions, aging, lighting and minor obstructions like glasses or masks. VGGFace offers a detailed representation of faces by converting images into the numeric vector that will be easily compared for similarity. In the Face Recognition System, VGGFace acts as the main engine for recognizing faces.

After detecting faces and preprocessing the images the system will uses vggface model that create feature embeddings for each face. Then compares these unique feature maps with stored embeddings in databases using similar measures such as cosine distance to find matches. This method helps the system identify criminals or locate missing children by recognizing even subtle differences and similarities in facial features. VGGFace can manage changes in expression, aging and minor obstructions, ensuring reliable performance in real-world situations such as surveillance footage or photos taken under different conditions. By providing accurate facial identification VGGFace improves the speed, accuracy and overall effectiveness of system.

❖ **SciPy**

SciPy is a freely available Python tool for scientific and technical computing. SciPy offers various tools for tasks like linear algebra, optimization, statistics, signal processing and numerical computation. Many people use it data analysis machine learning and ai projects because of its efficient algorithms and easy works smoothly alongside other Python tools such as NumPy and Pandas. In the face recognition systems scipy mainly calculates cosine similarities between two facial feature vectors or embeddings. Cosine similarities how similarities between these two vectors are regardless of their size. This make a perfect for comparing facial embeddings created by deep learning models like VGGFace.

By working out a similarity score SciPy helps determine whether an input images matches a stored image in the database. This is crucial for accurately identifying criminals or finding missing children. Additionally SciPy is wide range of scientific functions provides flexibility for other computations, optimizations or statistical analyses needed to improve model performance or evaluate results. This makes it an essential tool in the system.

❖ **Numpy**

Numpy is python tool for crunching scientific data. Numpy supports multi-dimensional array and matrices offers many math tools that help handle numbers quick easily computation. It is popular in data analytics machine learning and ai due to its speed flexibility and easy plays good with many python tool like scipy and tensorflow. In the face recognition system numpy is crucial in managing and processing image data. It converts images into numerical arrays and matrices which ai systems such as vggface that learn from images can use feature extraction, classification. numpy also used for in reshaping image data performing vector calculations and efficiently handling batches of images ensuring the system processes multiple images quickly and accurately.

Its high computational efficiency for managing large datasets from cctv footage webcams or uploaded files. This supports real time processing and smooth operation of ai based face recognition models. By providing reliable framework for representing manipulating images numerically, numpy serves key foundation of entire system.

❖ **PIL**

PIL its updated version pillow a popular python tool that helps with image processing. pillow offers tools to open, manipulate and save images in different types like jpeg, png bmp and gif. It makes common work is working with pictures easier such as resizing cropping rotating and changing images from one colour

format to other modes like rgb and grayscale. pillow plays nicely with other python tools like numpy and tkinter which makes it suitable for both backend image processing and gui applications. In the face recognition system pil helps load and prepare images for deep learning models. It converts images to rgb mode resizes and crops them to the necessary dimensions and ensures they are compatible with numpy arrays and facial recognition models like vggface. This preprocessing for keeping consistency and accuracy in face detection and recognition.

PIL also handles image files efficiently from sources such as cctv footage webcams or uploaded photos. When used with gui libraries like tkinter it lets users upload and preview images interactively which improves the system is user experience. by simplifying image preprocessing and ensuring proper formatting for deep learning pipelines pil contributes significantly to enhancing the overall performance and reliability of the face recognition system.

❖ Tkinter

Tkinter is pythons a core library used to make windows, buttons and menus in python apps. Tkinter offers set tool to creating desktop application with basic parts like windows buttons labels text inputs file dialogs and menus. It is lightweight and easy use and works good with python. This make an option for building interactive applications. Developers design intuitive and easy on the eyes interfaces without needing much experience in GUI programming.

In the face recognition system tkinter used to build user interface to uploading and managing images of criminals or missing children. It lets users browse select and input images into the system without using command line operations. This makes the application simple and accessible for everyone. This makes the system accessible to non-technical users such as police officers, administrators, or concerned citizens. This interface allow real time interactions with system, showing recognition results, similarity scores along other useful details immediately. By offering an intuitive gui, Tkinter improves ease of use, operational efficiency and accessibility. This ensures the system can be effectively used in real world situations like law enforcement investigations or missing child searches. Its work together with tools like pil and opencv allows for smooth image handling and preprocessing which is essential for accurate face detection and recognition.

❖ OS, Shutil and Sys Modules

OS, Shutil and Sys are built in python modules that manage system level operations and interact with the operating system. The os module provides tools to access and manage the file system. It allows users

create delete and navigate directories. Additionally it offers access to file information. shutil is a high level module that helps copy move and delete files or directories effectively. The Sys module allows interaction with the python runtime environment. It manages system specific settings command-line arguments and program identification. Together these parts are key to handling files directories and system tasks in python applications. In the face recognition system they important automate background tasks and managing project data. The os module enables the system to access image directories organize datasets and navigate through large numbers of uploaded images. shutil assists in copying moving or deleting files which is particularly useful for managing photos of criminals or missing children. The sys module ensures smooth management of runtime parameters and program exit. by combining these modules the system can automate file handling keep a tidy directory structure and reliably save and retrieve images. These steps are key to the smooth operation and high performance of the face recognition pipeline.

- **Matplotlib**

Matplotlib is python library that works with data visualization. Matplotlib offers wide range of tool to creating graphs charts histograms and plots. It also allows displaying modifying images. The library supports various visualization styles formats which makes it useful for both simple and complex visualizations in scientific computing machine learning and data analysis. Matplotlib plays nicely with other python tools like numpy, scipy and opencv enabling developers to visualize numerical and image data effectively.

In the face recognition system matplotlib displays input images detected faces and recognition results. It shows face detection bounding boxes that pinpoint the exact location of the detected faces. When methods like grad cam are used. matplotlib can present heatmaps or visual explanations helping users understand the reasons behind model recognition.

By providing side by side comparisons input and matched images matplotlib improves clarity usability. This makes easier verify and interpret recognition results. it boosts interactivity, transparency and informativeness in the systems. Users can visually confirm face detection and recognition outcomes in a clear and intuitive manner.

- **VS Code**

Visual studio code is free open source and very flexible code editor created by microsoft. vs code works with lots of programming languages including python, javascript and c++, making it suitable for wide large range development projects. It comes with feature such as intellisense for smart code completion debug

tools an integrated terminal syntax highlighting and git version control integration. Users can add extensions for formatting and linting and jupyter notebook support and development tasks which improve productivity coding efficiency.

In the face recognition system vscode serves the main environment is for writing and editing and debug python scripts. It helps developer organize project files manage libraries run code smoothly making it easy to integrate tool like openccv, mtcnn and vggface, scipy numpy, pil and tkinter. The built in git support proper version control and collaboration which is essential for keeping code consistent and tracking changes during development process.

Additionally vscode is debugging tool and live error detection allows developer identify fix issues quickly reducing development time. It customize interface support for extensions and has ability to manage virtual environment help to grow more smoothly and maintains in face recognition application. Overall vscode enhances productivity and efficiency and reliability providing a strong platform for developing testing and deploying entire system easily.

2.4 Hardware And Software Requirements

❖ **Hardware Requirements**

Requirement	Specification
System	Intel I3 processor
Hard Disk	500GB
RAM	4/8 GB

❖ **Software Requirements**

Requirement	Specification
Operating systems	Windows 8 or above
Programming languages	Python 3.7 or Above
Software	Visual Studio Code, Jupyter Notebook

3 SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Users

Users are people or organizations who engage with the system to identify individuals or solve related cases. They use it to quickly find matches and get good results. This leads to faster processing, less manual work and better decision-making.

❖ **Law Enforcement Officers**

Police officers, detectives and additional police or security team members are the main users of this system. They depend on the software to identify criminals and find missing children by uploading facial pictures coming from different places like CCTV cameras, crime scene photos or public alerts. The face recognition system analyses these images in real time and compares them to the existing database. This significantly cuts down the requirement for manual investigation.

The System built to handle real world situations. It can manage age progression, disguises, changes in facial appearance and low-quality or partially covered images. It ensures high accuracy even in challenging conditions. By providing quick identification results it helps law enforcement officers respond promptly, make informed choices and focus on urgent cases.

Overall system improves the efficiency of criminal investigations and searches for missing persons. It lightens the operational workload and boosts public safety by enabling faster identification of suspects and quicker recovery of missing individuals. Is easy to use interface allows officers with limited technical skills to operate the system smoothly turning it into a handy tool for modern law enforcement.

❖ **Social Service Departments and NGOs**

Organizations that focus on child protection, preventing human trafficking, and social rehabilitation are important users of the system. Caseworkers and social service staff can upload photos of missing or unidentified children inside the setup. It then compares image to the facial database to find possible matches. This process helps reunite families quickly, easing the emotional and administrative burdens on both the children and their families.

The system is also helpful in identifying abandoned or unidentified children. It supports shelter placement, adoption programs and social reintegration efforts. By offering real time results through an easy to use interface it enables NGOs and social service departments to respond quickly in sensitive situations where delays could have serious effects. The technology allows caseworkers to access useful information without

needing advanced technical skills. This improves operational efficiency, increases child safety and supports broader social welfare goals.

❖ **Government and Intelligence Agencies**

Agencies tasked with national security, border control and public surveillance are key users of the face recognition system. These agencies use the software to monitor, track and identify persons of interest in crowded spots such as airports and train stations, government buildings or public events. The system employs face scanning and matching technologies to compare individuals against national identity databases, criminal records and watchlists. This helps identify wanted criminals or locate missing persons across different jurisdictions.

By connecting with national and regional database the system fosters collaboration among agencies. This allows intelligence and law enforcement to share information effectively. It improves information gathering, boosts threat detection and supports quick responses in critical situations such as preventing crimes or finding vulnerable individuals. Additionally the system provides accuracy and speed in monitoring activities. It lessens the manual tasks for security staff while upholding public safety and national security. Its use enables agencies to respond to potential threats more proactively and enhances the overall effectiveness of surveillance and security operations.

❖ **General Public**

A simplified and secure version face recognition system can be available to THIS public with proper authentication and data privacy measures. This version allows family members or concerned citizens to report missing people by uploading photographs. The system then compares these images against an open access missing persons database. If it finds a potential match, authorities are notified which helps reunite families more quickly and efficiently.

Giving the public access to the system encourages community involvement in finding missing people. This creates a network between citizens and law enforcement or social service agencies. Strict data privacy and verification protocols ensure that uploaded information is used ethically and safely protecting both the public and the individuals in the database. By enabling responsible public engagement the system not only speeds up the identification and recovery process but also raises awareness and involvement in missing person cases turning it into something really useful for everyone.

3.2 Functional Requirements

❖ **Face Recognition Capability**

- The System will be able to identify criminals or missing children accurately even with significant changes in appearance or aging over time.
- It should use a strong deep learning model that can recognize faces in different lighting conditions, facial expressions or partial obstructions.
- The recognition process should provide consistent and reliable results while minimizing false positives and negatives.

❖ **Model Training and Performance Evaluation**

- During training the system should show detailed performance metrics, including accuracy, loss, precision, recall and validation results.
- These metrics help developers evaluate how well the model works, adjust parameters and make sure it can generalize to new or unseen data.
- The system should allow for updates or retraining of the model when new data like additional images of criminals or missing children is available.

❖ **Image Handling and Processing**

- The system must let users upload images in various formats such as JPG, PNG and BMP to support different sources.
- It should process images of different resolutions and sizes automatically resizing and preparing them for accurate recognition.
- The system must support batch processing so multiple images can be processed efficiently without manual input.
- Preprocessing features should include converting to grayscale, cropping faces and normalization to ensure consistency for deep learning models.

❖ **User Interface and Usability**

- The system should offer a simple, interactive and user-friendly interface for users with limited technical knowledge.
- The interface should display matched results, similarity scores and bounding boxes around detected faces for clarity.
- It should be designed for law enforcement officers, social workers or authorized personnel to ensure practical use in real world investigations or emergencies

3.3 Non-Functional Requirements

❖ **Scalability and Robustness**

- The system must recognize people even when their facial features are partially obscured such as with blocked faces or low-quality images.
- It should manage different image qualities and minor obstructions ensuring reliable recognition in real world situations.
- The system should scale well handling more images, a larger database and more users at the same time without slowing down.

❖ **Hardware and Software Compatibility**

- Users must meet all basic hardware requirements, including enough RAM, GPU support and storage to guarantee smooth processing.
- The system should work with essential software libraries, dependencies and frameworks like Python, OpenCV, TensorFlow and Keras.

❖ **Responsiveness and Performance**

- The system should deliver quick results typically within a few seconds to support real-time applications such as surveillance or fast identity checks.
- The recognition process must be effective using optimized algorithms and hardware acceleration when possible to reduce delays.

❖ **User Interface and Usability**

- The system must have user friendly interface that individuals with limited technical skills can easily use.
- It should guide users through uploading images, recognizing them and interpreting results, making the whole process clear and easy.
- The interface should provide consistent functionality across various platforms and devices to ensure a smooth user experience.

❖ **Reliability and Stability**

- The application must be strong and dependable, avoiding crashes, freezes or unexpected behaviour during use.
- It should manage exceptions well, offering clear error messages or backup options when problems arise.

4 SYSTEM DESIGN

4.1 System Perspective

The proposed system uses a convolutional networks based design called AIFR cnn to recognize faces of missing people and criminals. The process converts 2D facial images into high dimensional feature representations. In this space features from the same individual come together while features from different individuals stay apart, which helps with accurate identification.

Face detection starts with a camera capturing raw images that contain both background and face regions. A CNN-based face detection algorithm identifies and isolates the facial regions within these images. At this stage, the system separates facial patterns from non-facial elements.

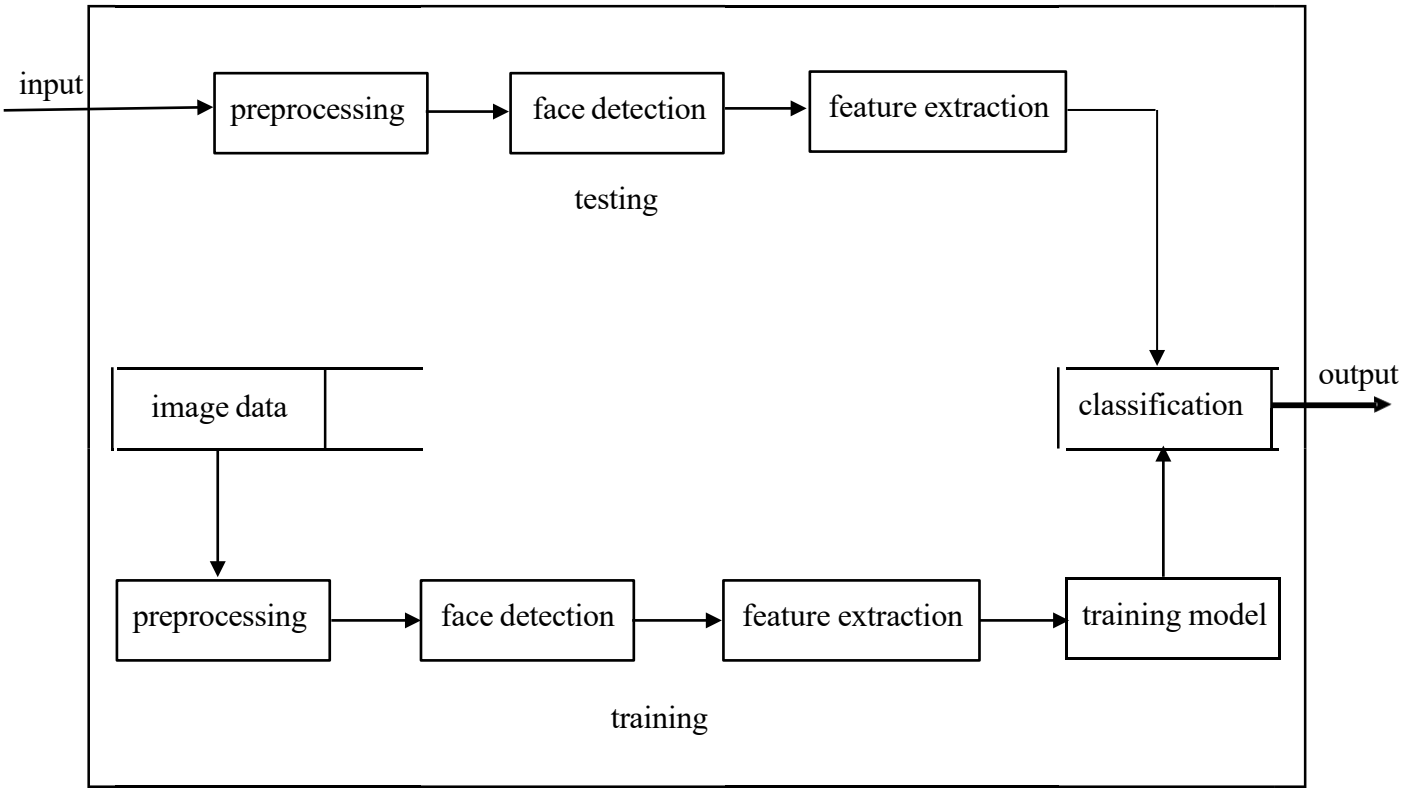


Fig 4.1: System Perspective of Deep Vision – AI for missing children and criminal investigation

Once detected the face image goes through normalization. This includes cropping the facial region resizing the image converting it from rgb to grayscale applying histogram equalization. This improves contrast and brightness to handle different lighting conditions making sure image quality consistent for analysis.

These methods pull out relevant facial descriptors without manual help. The extracted features move through a 7 ayer CNN architecture for classification. The system must identify individuals from a large group it is treated multiple class classification problem.

4.2 Context Diagram

This is high level data flow diagram. It represents one process and shows how it interacts with external entities. It focuses on inputs outputs external interfaces without getting into internal details. for your project context diagram shows users uploading images the ai system processing them and system returning results.

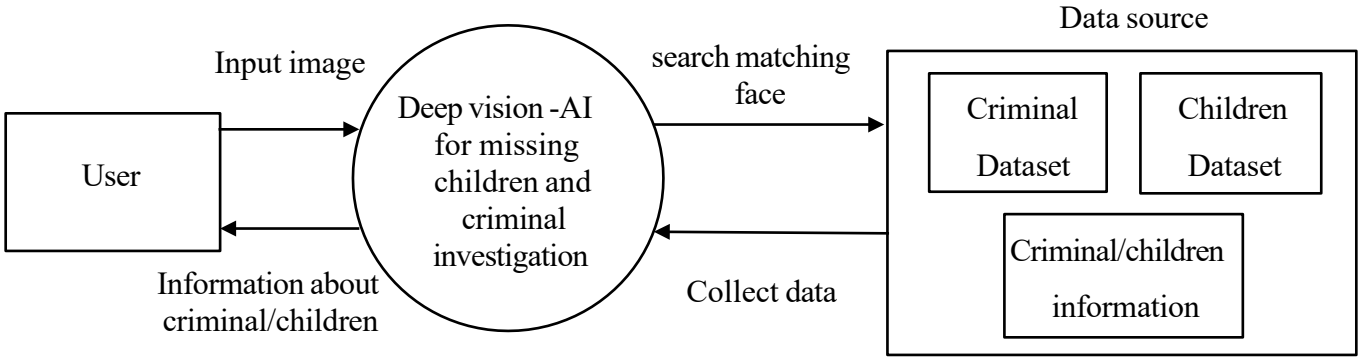


Fig4.2: Context Diagram of Deep Vision – AI for missing children and criminal investigation

The image shows a context-level data flow diagram for an AI based system that helps in searching for missing children and in criminal investigations. This diagram visually outlines how the system works with outside sources like users and data sources. It specifically demonstrates how a user inputs an image which the AI system processes. The system communicates with various data sources including criminal image datasets, missing children datasets and information databases about children and criminals by searching for matching faces and gathering relevant data. The resulting information is then provided back to the user.

❖ Level 1 Data flow diagram

This level breaks down the main system process into sub-processes. It illustrate information flows them sub processes and external entities. For your project it might display separate modules for image upload, preprocessing, feature extraction, face recognition and result display.

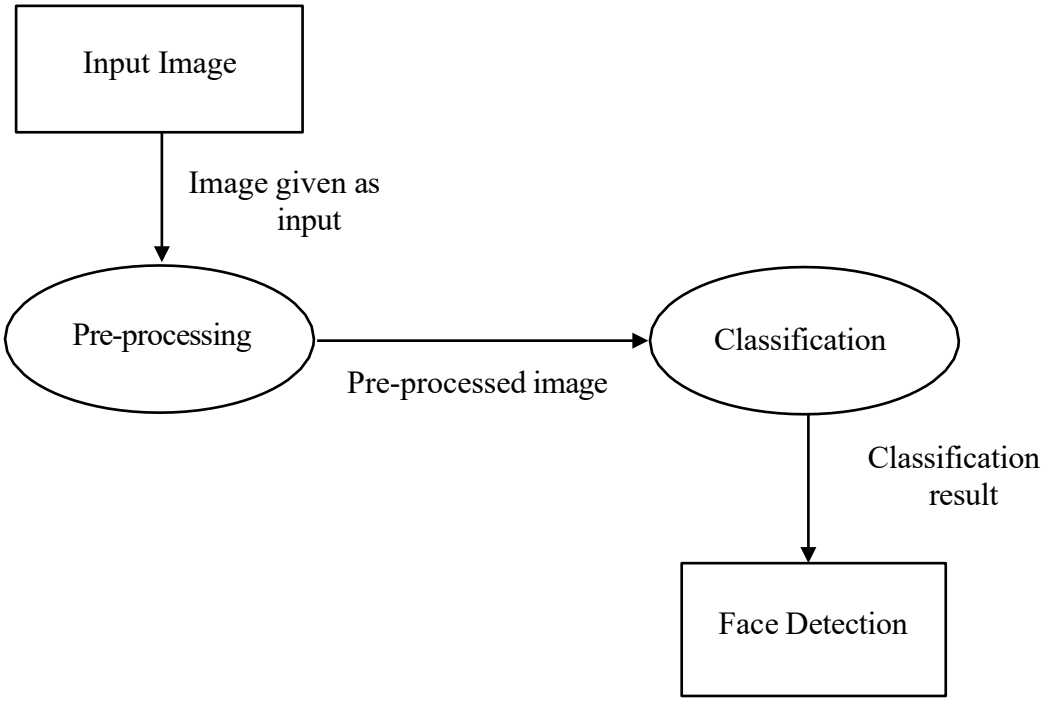


Fig4.3: Level 1Data Flow Diagram of Deep Vision – AI for missing children and criminal investigation

This stage offers a general overview of face recognition. It starts with an Input Image that first goes through a Preprocessing stage. As noted this preprocessing includes converting the RGB image to grayscale to simplify it followed by noise removal, applying Haar Cascade and resizing. The output of this stage is a Pre-processed image, which then moves into a Classification module. This module ultimately produces the Face recognition result. This flow emphasizes the separate steps of preprocessing and classification.

❖ Level 2 Data flow diagram

Level2 DFD offers a more detailed breakdown of a specific Level 1 process. It illustrates the internal workings of individual subprocesses, such as how the face recognition module performs MTCNN detection, feature extraction with VGGFace and similarity matching with SciPy. It also includes data flow between each step.

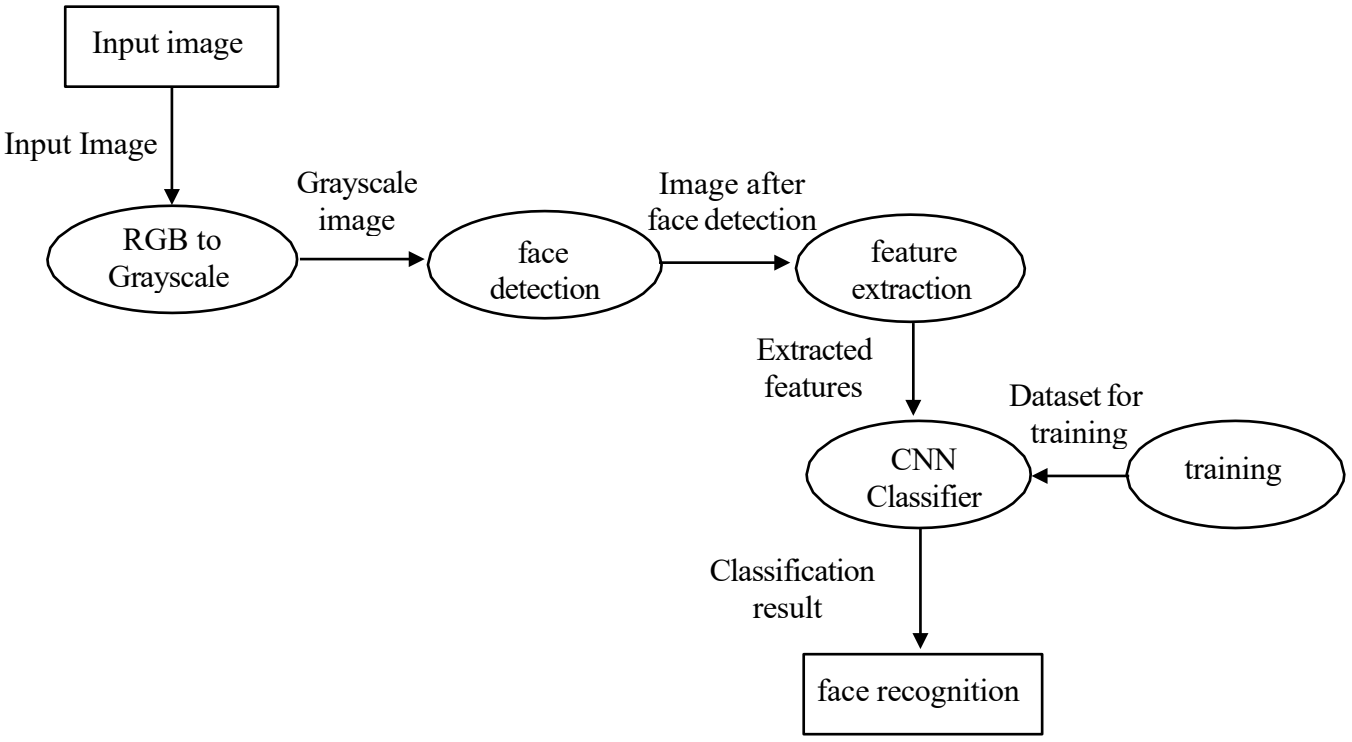


Fig 4.4: Level2 Data flow diagram of Deep Vision – AI for missing children and criminal investigation

This diagram shows a flow of data that follows CNN for face recognition. The process starts with an input image which is converted from RGB to grayscale. The grayscale image then used face detection resulting in an isolated face image. This detected face is processed to extract feature which are sent to a CNN classifier.

During training the CNN uses a dataset for supervised learning to find patterns in the features. Once trained. the CNN produces a classification result, enabling accurate face recognition.

It is an important note CNNs can often extract features on their own. In some cases a pre-processed image can be directly input into the CNN for classification. Compared to a typical data flow that only involves pre-processing, this approach highlights more detailed steps. It shows how CNNs simplify feature extraction and improve recognition accuracy.

5 DETAILED DESIGN

5.1 Use Case Diagram

The core flow of system, shown in the diagram and discussed in text starts with Input. This means the images captured initially are temporarily stored in OpenCV. Next a Pre-processing step is essential where the snapped RGB images are changed into another format into grayscale to make the task easier and additional preprocessing techniques are applied. After that feature extraction is done on these pre processed images. After that the important details are used train system model phase where classifier trained available data to accurately detect faces. Classification follows using these trained classifiers to identify faces. Finally, the system carries out face recognition. The end result simply labelled Result shows that the detected face is displayed along with intermediate results and various information about the recognized face.

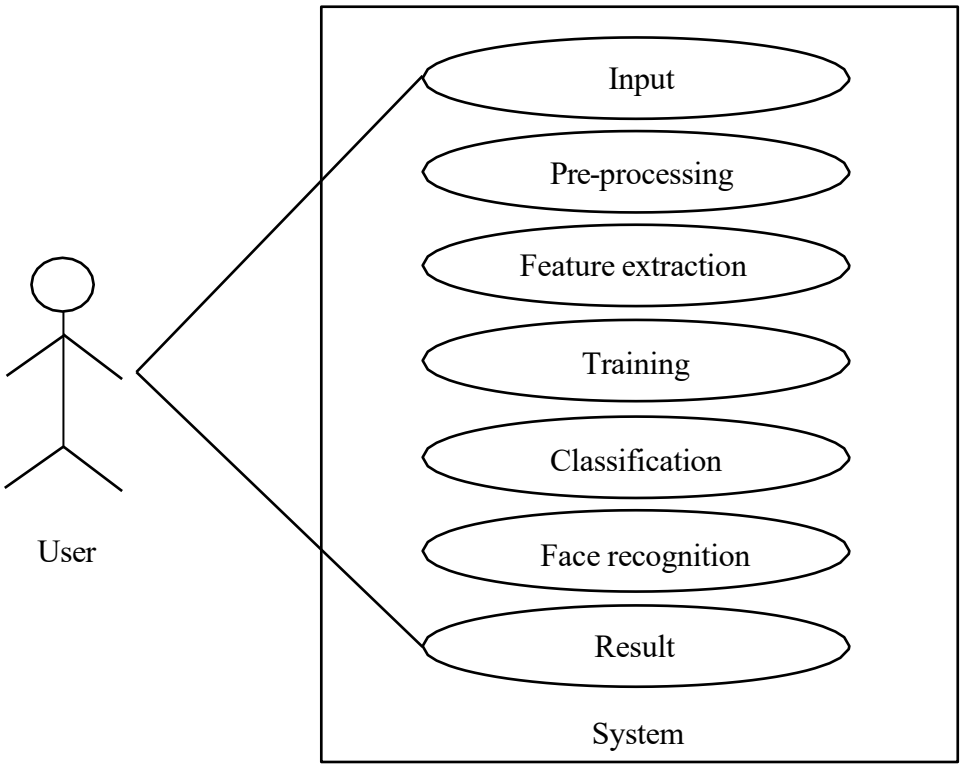


Fig 5.1: Use Case Diagram of Deep Vision – AI for missing children and criminal investigation

The use case diagram visually shows how the User interacts with the Input to start the process and how the System through a series of internal use cases ultimately provides a Result back to the User. This entire process is focused on the specific application of identifying missing criminals and children.

5.2 Sequence Diagrams

A sequence diagram illustrates the order of shows how system pieces communicate during time. In this project it shows the steps from image upload, pre-processing, face recognition to result display. It explains how each part talks to the others to complete the process.

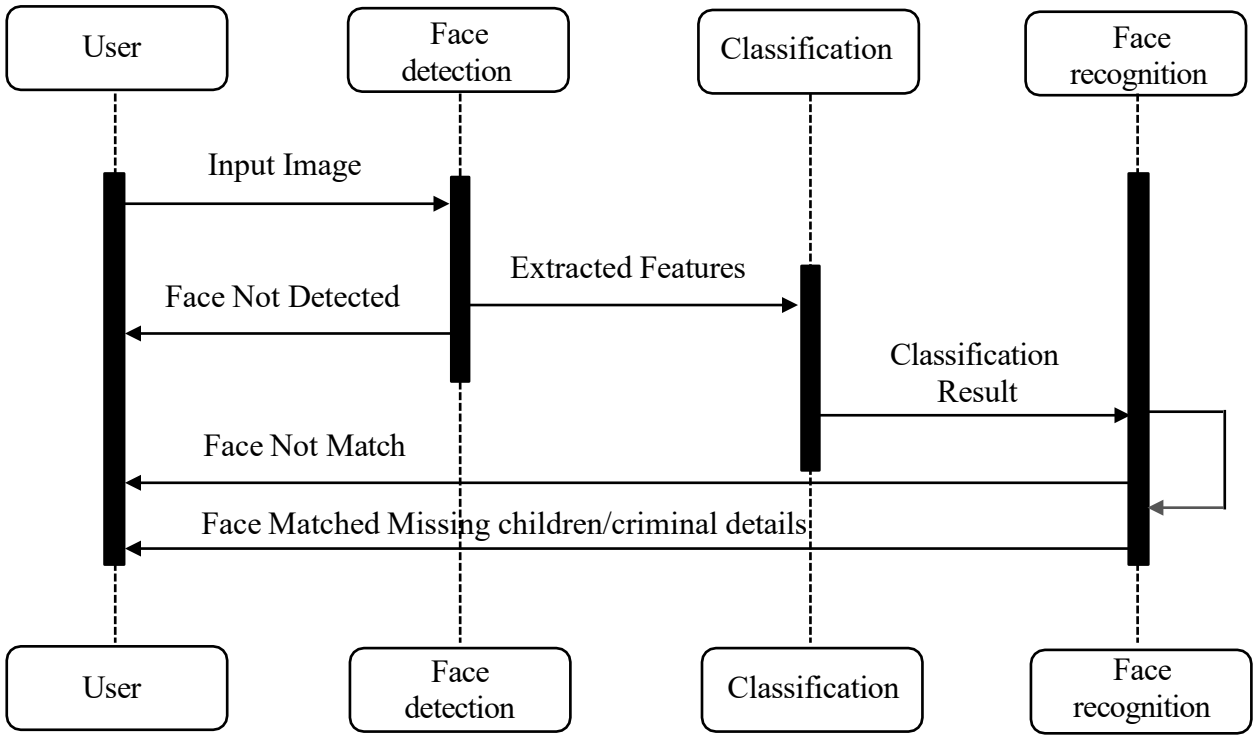


Fig 5.2: Sequence Diagram of Deep Vision – AI for missing children and criminal investigation

This sequence diagram outlines the flow of interactions within a face recognition system. The User starts the process by submitting an Input Image to Face Detection. Face Detection then checks if a face is present either returning Face Not Detected to the User or passing Extracted Features to Classification. Classification processes these features and sends a Classification Result to Face Recognition. At last Face Recognition tries to match the face either informing the User of a Face Not Match or providing Face Matched Missing Children/Criminal Details upon successful identification.

5.3 Collaboration Diagram

A collaboration diagram emphasizes how objects are organized and interact with one another. In this project, it represents the relationship between the GUI image processing module, AI model, and databases. It shows how they exchange messages to perform recognition.

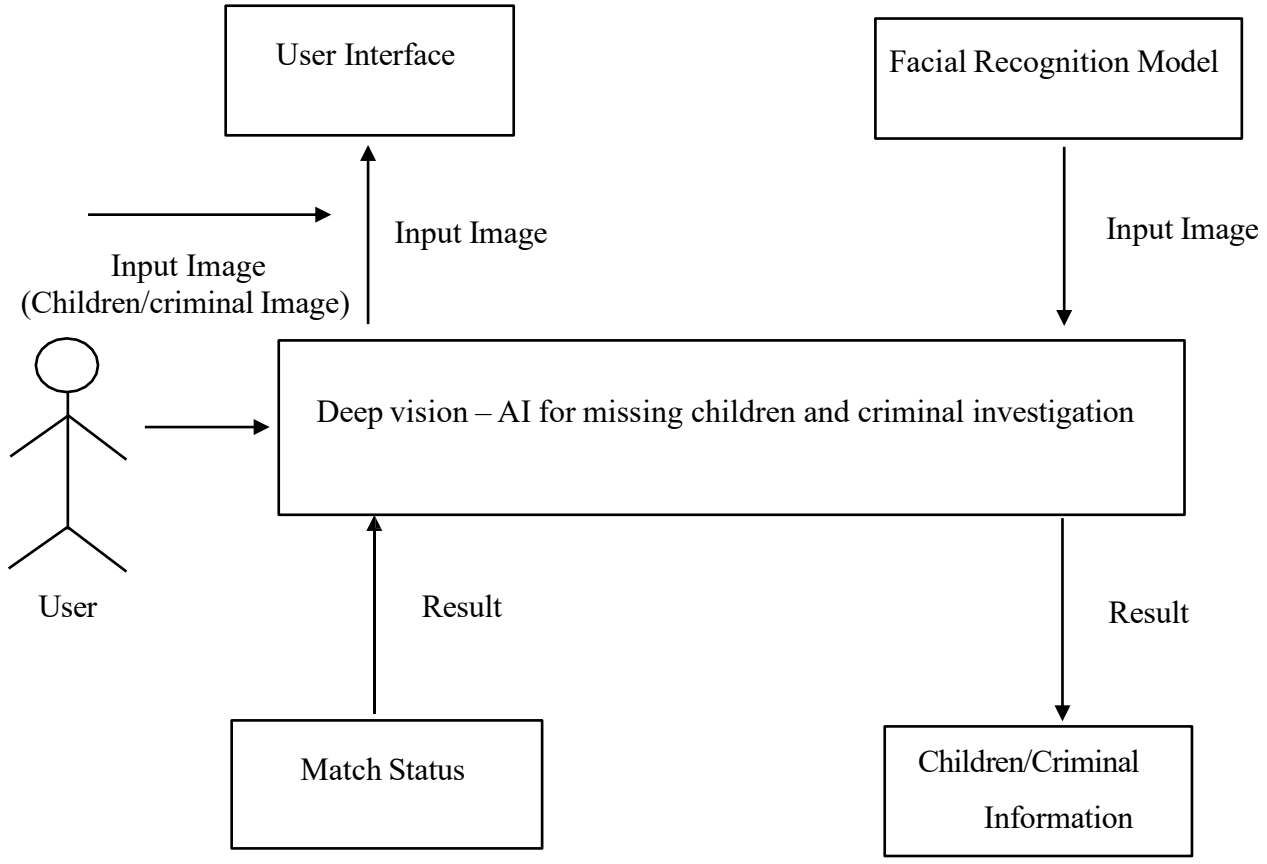


Fig 5.3: Collaboration Diagram of Deep Vision – AI for missing children and criminal investigation

A collaboration diagram shows how objects or components in a system interact. It focuses on their relationships and the messages exchanged to complete a task. The diagram illustrates how different parts of the system work together detailing the roles of objects, their connections and the sequence of communication. It really helps to see how components cooperate, ensuring messages flow efficiently and confirming that each object performs its role in meeting system goals.

5.4 Activity Diagram

An activity diagram models the workflow of tasks in the system. In this project, it describes the process where the user uploads an image. The system detects the face, extracts features compare them with the database and displays results. It includes decisions like whether a face is detected or a match is found.

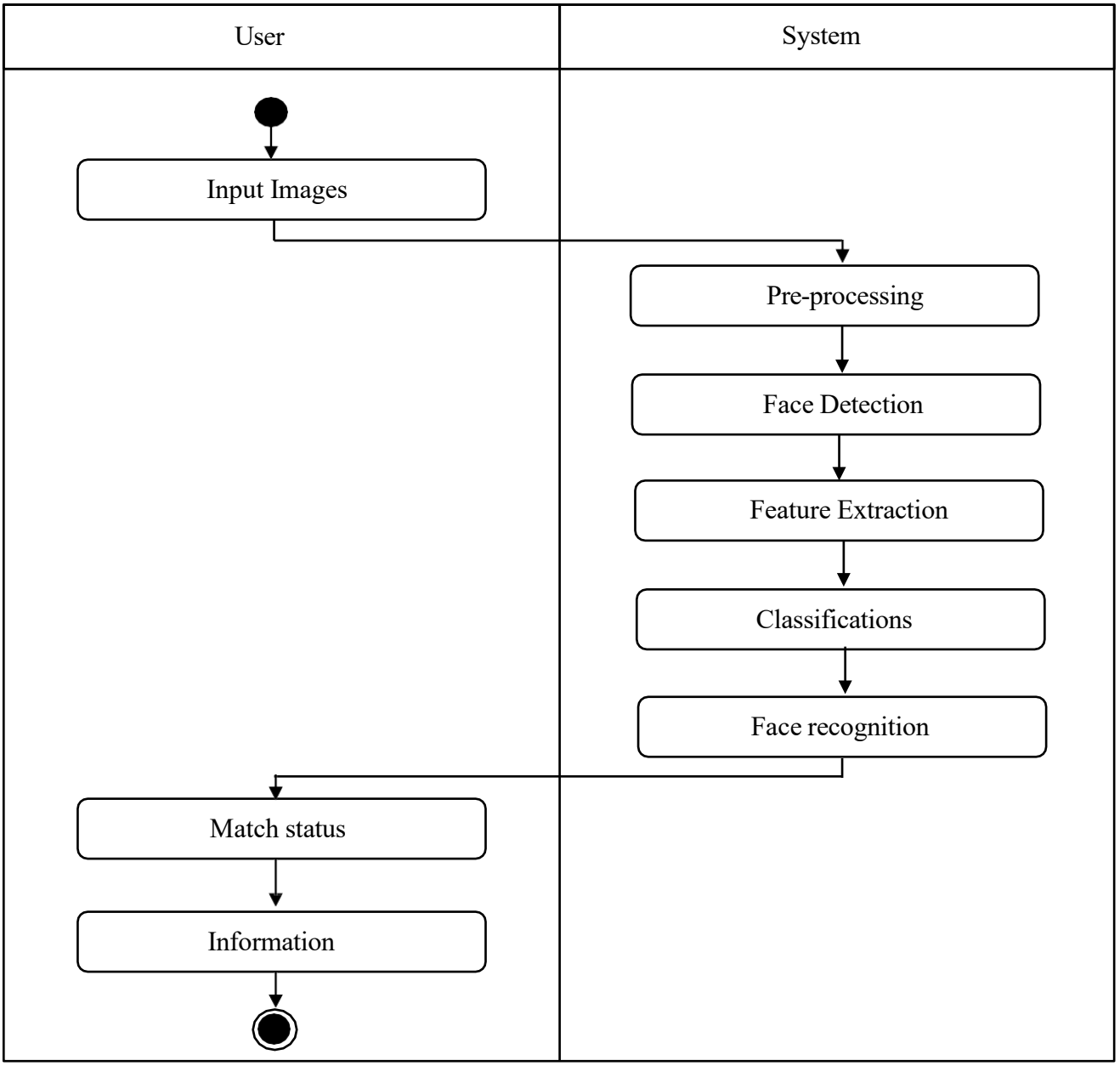


Fig 5.4: Activity Diagram of Deep Vision – AI for missing children and criminal investigation

An activity diagram outlines the flow of activities and actions within a system. It highlights the sequence of operations, decision points and processes that occur at the same time. The diagram is useful for modelling workflows, system functions or use case processes. It helps developers and stakeholders grasp system behaviour, process potential outcomes, making sure the tasks get done smoothly and efficiently.

6 IMPLEMENTATION

6.1 Screen Shots

❖ **Home page for the Deep Vision – AI for missing children and criminal investigation**

The main page serves as the main interface where users begin. It lets users choose an existing image or upload a new one for analysis. This platform is easy to use and helps process images to get good face recognition results quickly.

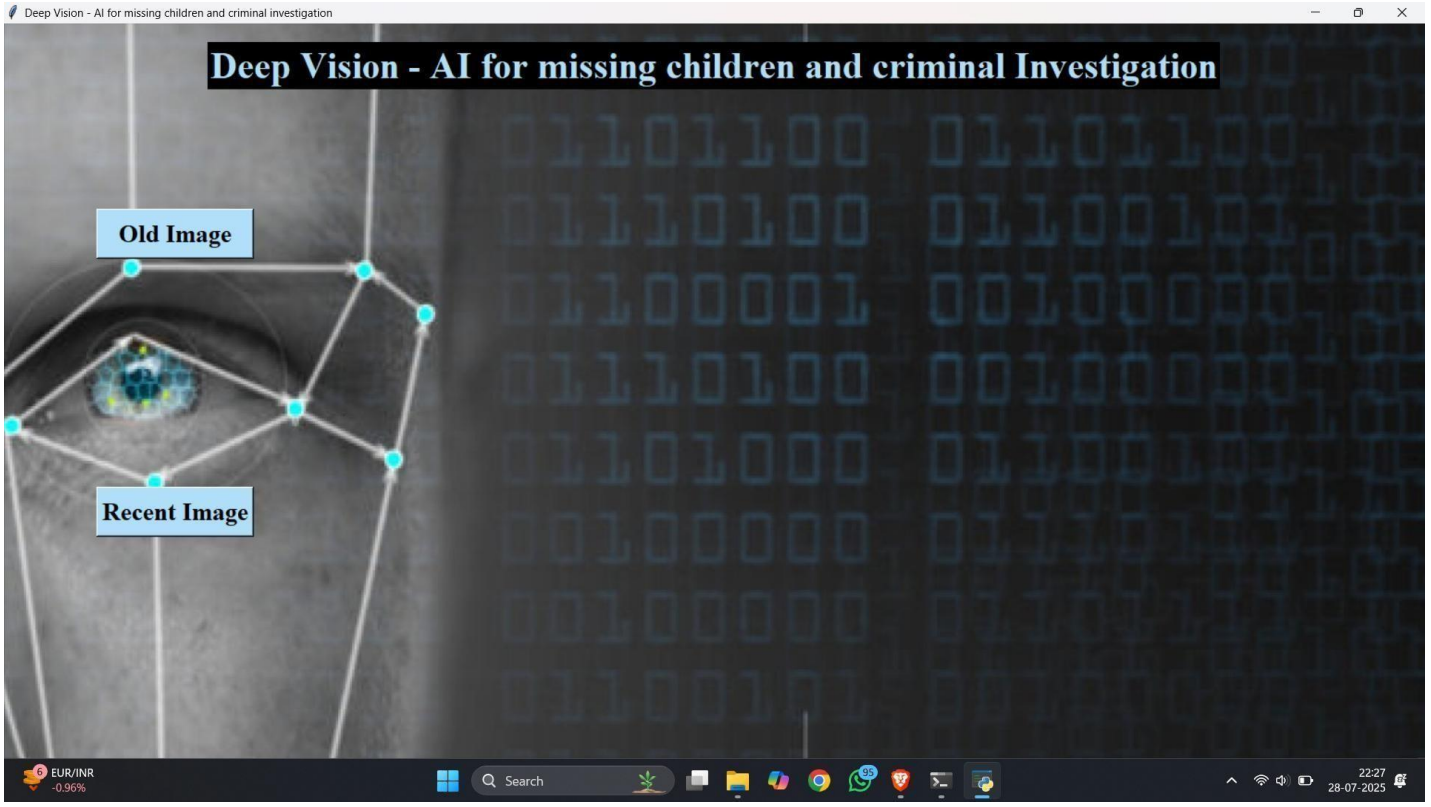


Fig 6.1: Home page for the Deep Vision – AI for missing children and criminal investigation

Users can choose an existing image either pick one from the database or add a new image for analysis. This lets the system process the image and deliver precise face recognition results quickly.

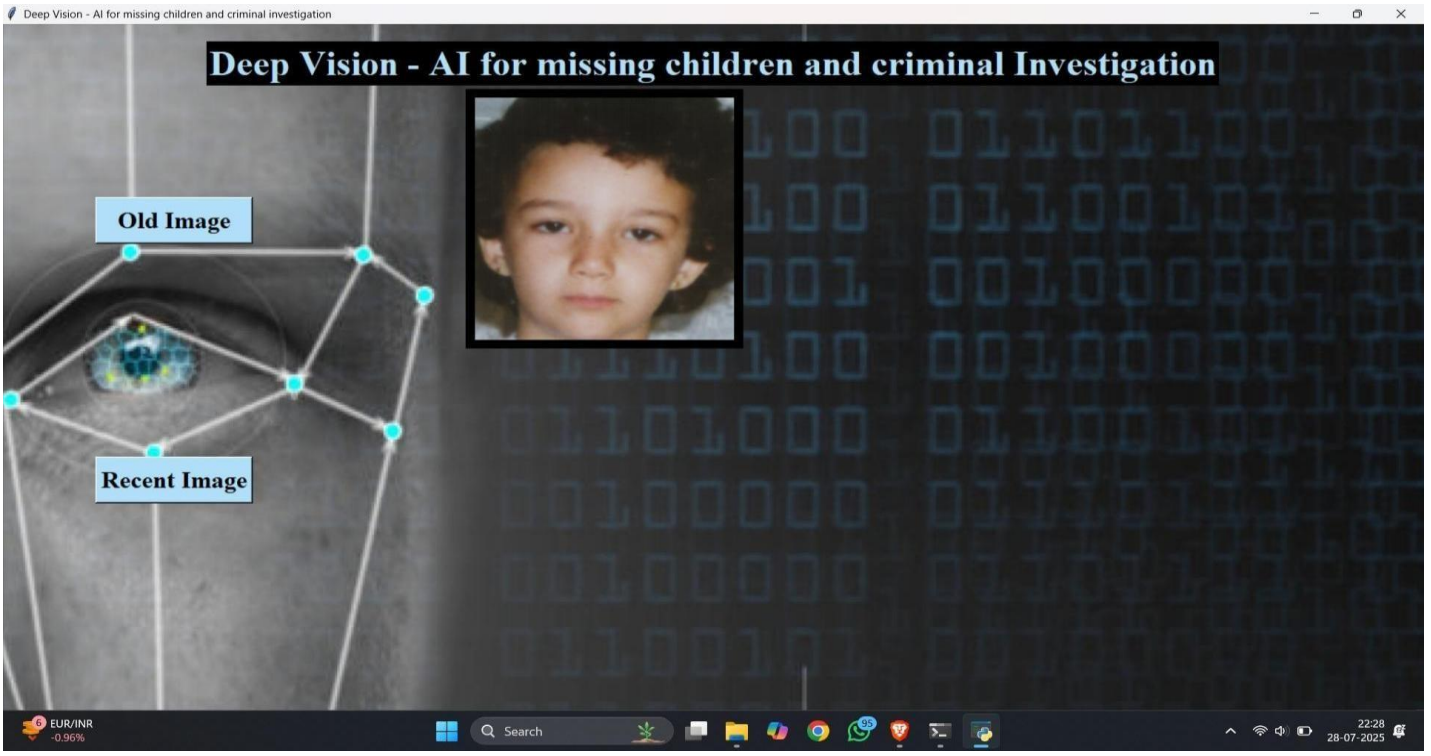


Fig 6.2: Upload old image to the Deep Vision – AI for missing children and criminal investigation

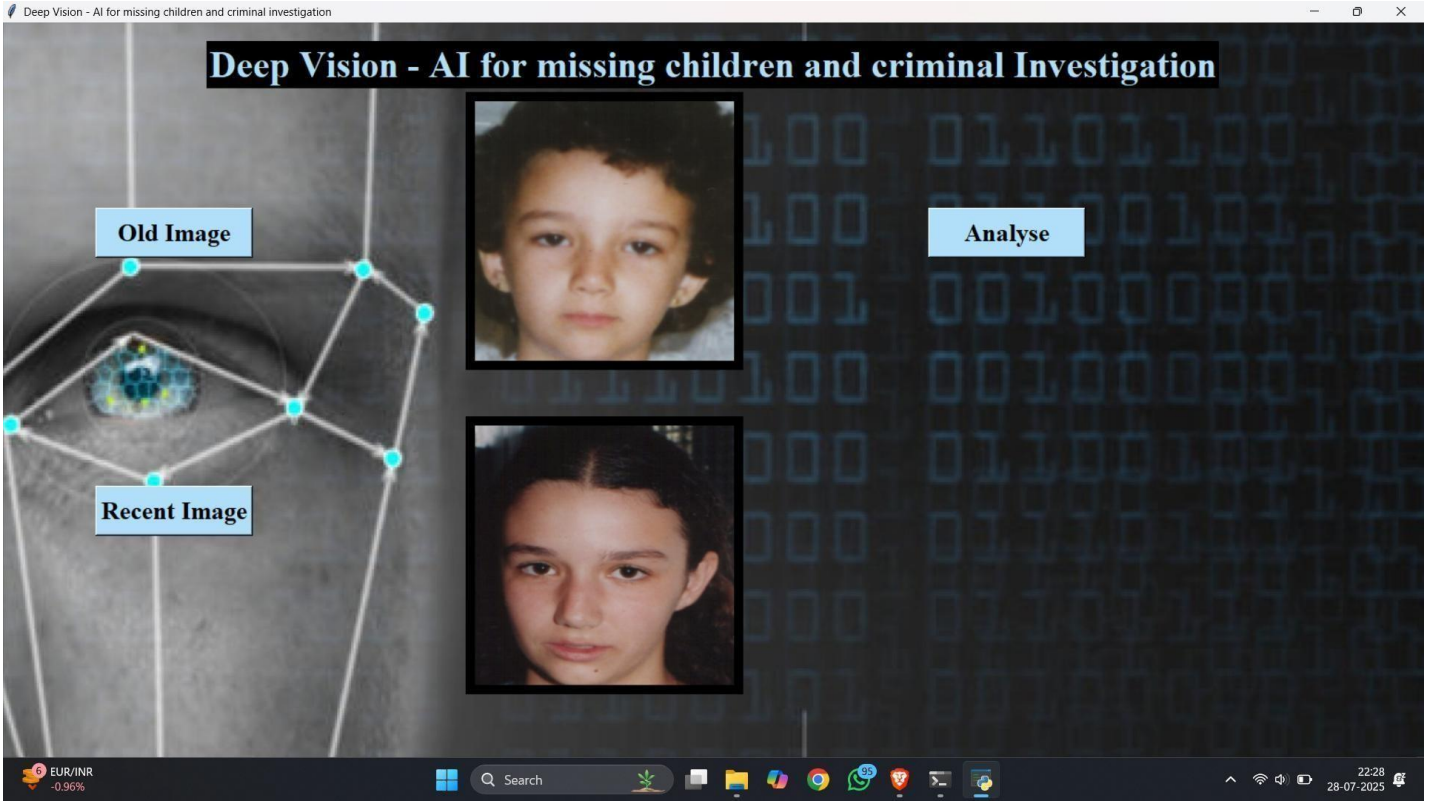


Fig 6.3: Upload recent image to the Deep Vision – AI for missing children and criminal investigation

Users begin at home page they can add images, start analysis and view results. The interface is straightforward making it easy to quickly identify missing children and criminals.

❖ **Child details**

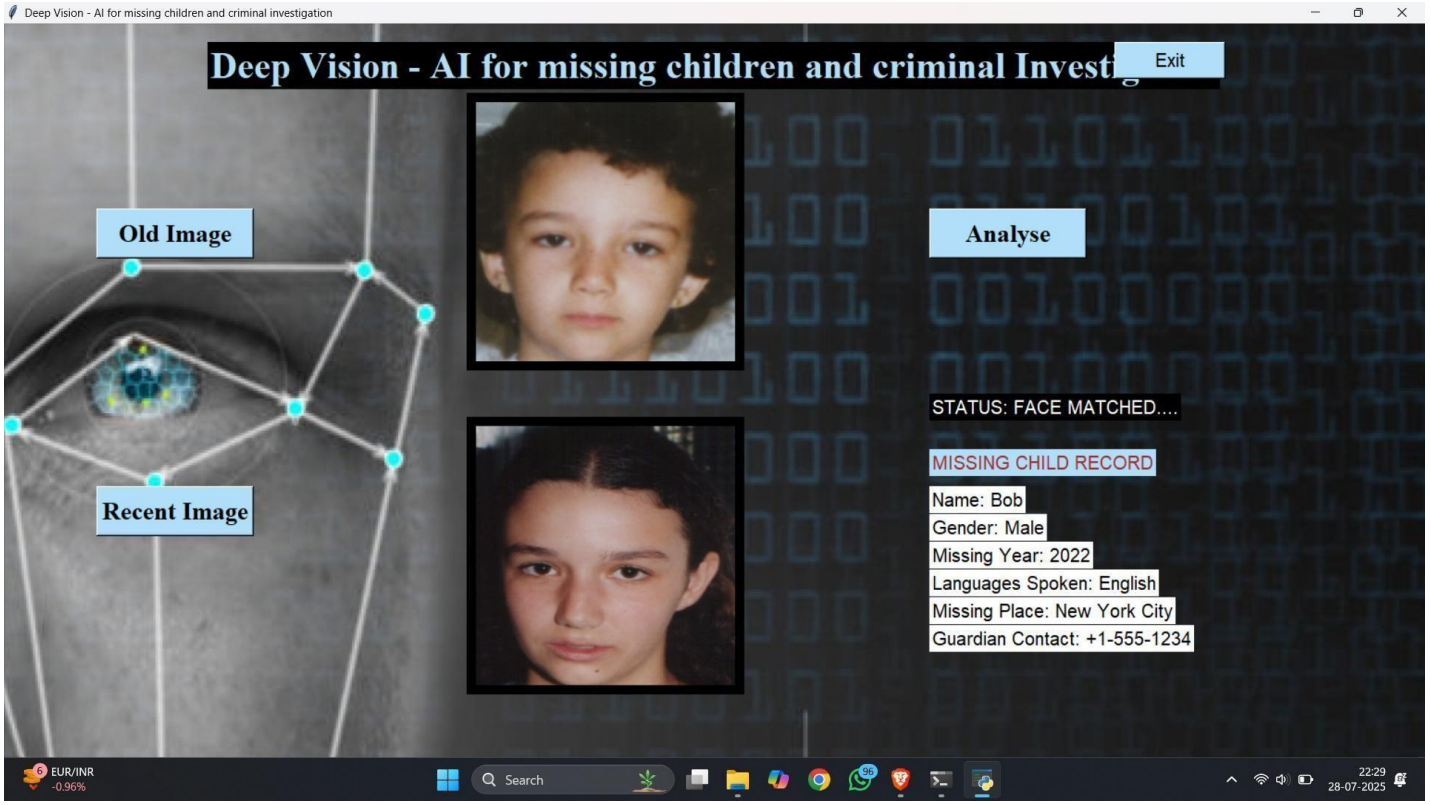


Fig 6.4: Identifying child by using Deep Vision – AI for missing children and criminal investigation

This section helps users find missing children from uploaded images. It shows important details like name, age, year missing, last known location and guardian contact information. This allows for a quick and informed response.

❖ **Criminal details**

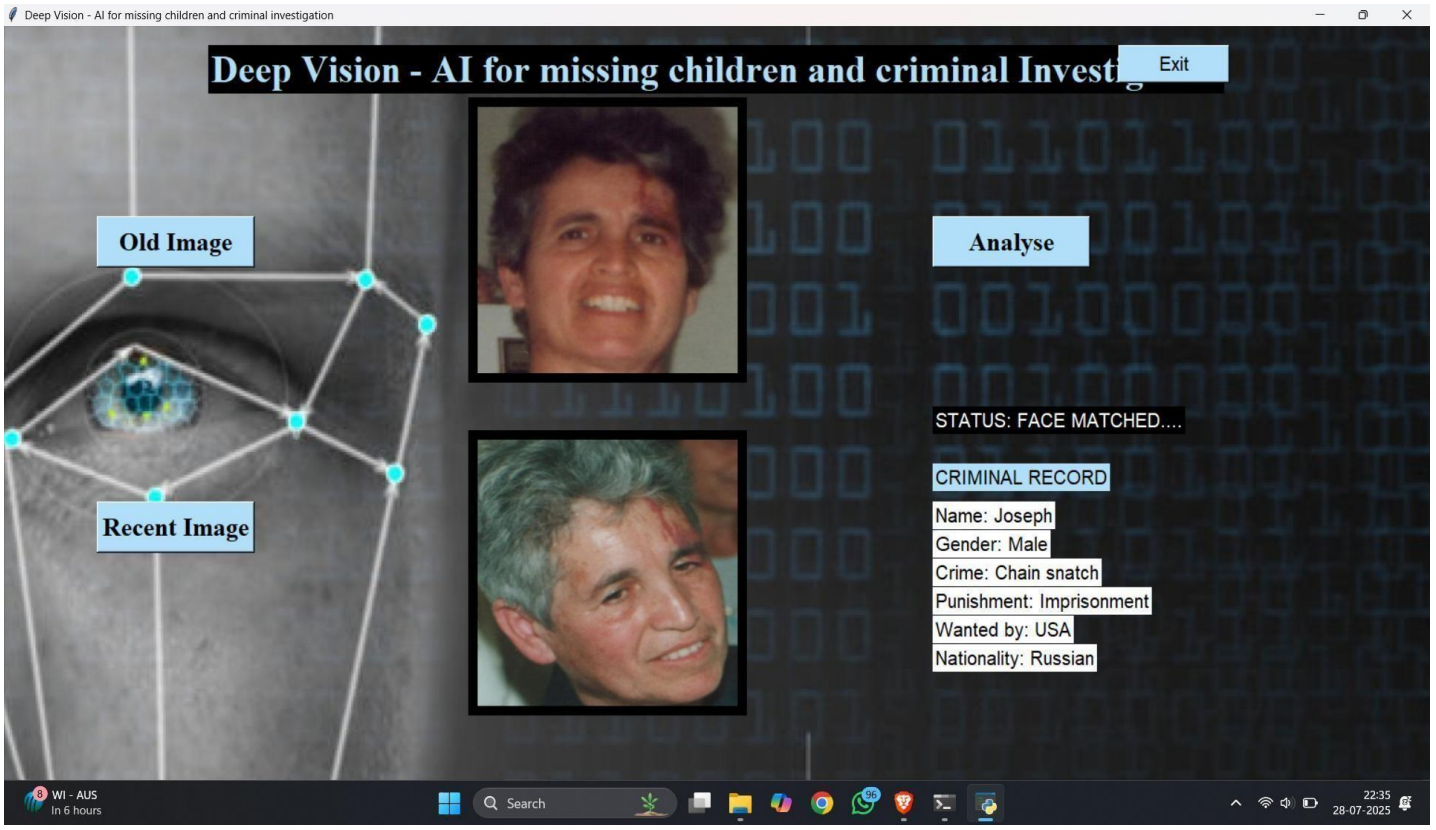


Fig 6.5: Identifying criminal by the Deep Vision – AI for missing children and criminal investigation

The criminal identification page shows matched faces along with important details. This includes the name, crime committed, punishment, nationality and wanted status. It gives authorities a quick reference to help with investigations and fast decision making.

❖ **Face not matched**

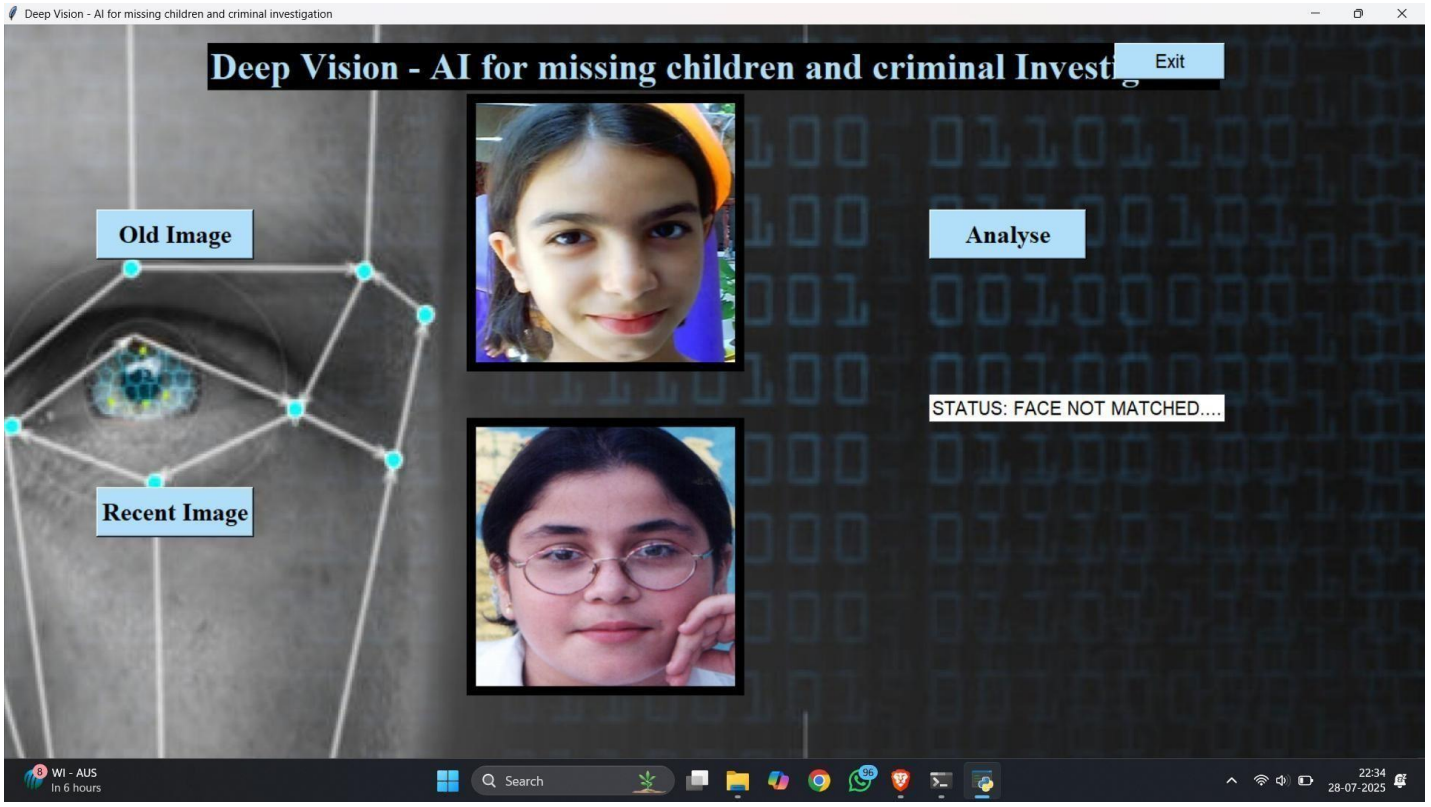


Fig 6.6: The face not match

If the uploaded face not match any records in database. It shows that the individual is not currently listed as a missing child or a criminal. This result helps users understand that there are no existing records associated with the submitted image ensuring reliable verification.

❖ **Face matched but no records found**



Fig 6.7: Face match but no records found

The uploaded face has been successfully matched in the system, but no corresponding records exist in the database. This means that the individual is not listed as a missing child or a criminal. Users can be sure that the person is currently unregistered in official records.

❖ **Face not detected**

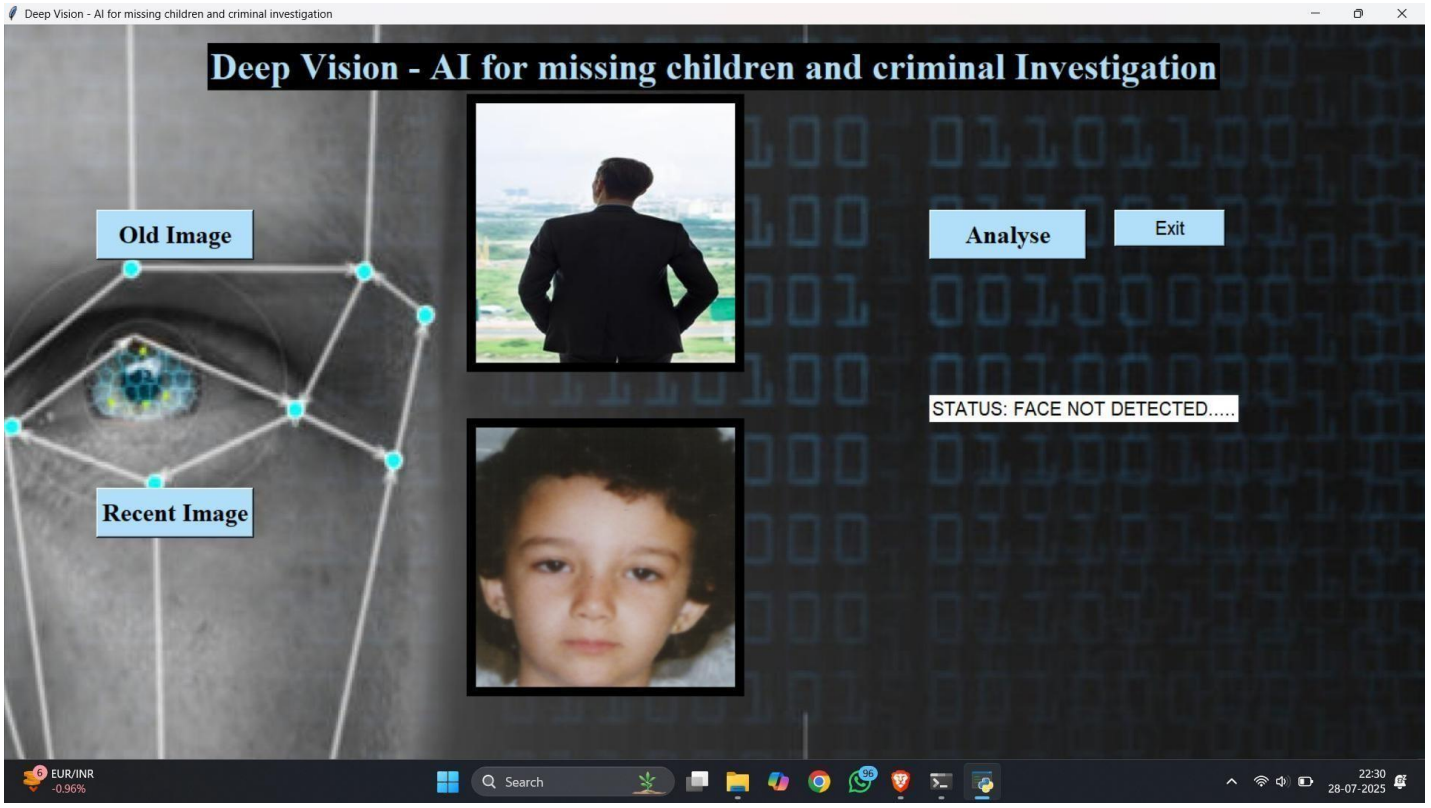


Fig 6.8: Face not detected.

The system couldn't spot any faces in this image. To ensure clear analysis and reliable results please provide a clear, front facing image where the face is fully visible. This will allow the system to properly process and match it against the database.

7 SOFTWARE TESTING

7.1 TESTING APPROACHES

❖ **Unit Testing**

For the Missing Children and Criminal Identification System, unit testing checks whether each function or component works correctly and independently. this methodology ensures every part of the system, including image upload, face detection with MTCNN, feature extraction and classification logic performs as expected on its own.

Each unit, like image pre-processing, face cropping and the matching function, is tested separately to catch potential issues early development process. Unit testing helps find problems like incorrect face embeddings, unsupported file formats, image resolution errors or processing failures before the components merge into the complete system.

By testing each module thoroughly, developers can make sure each function is reliable, correct and efficient. This practice reduces errors in the final system and improves overall stability and performance during real time use.

❖ **Integration Testing**

Integration testing check different part of the systems work together as one. In this project focuses on user interface image processing tools deep learning model coordinate with each other.

The testing makes sure this data flows across all components. for example when user uploads an image through the interface the system must process the image correctly extract facial features and display accurate recognition results. Integration testing finds problems in communication between modules such as mismatched data formats processing errors or delays in displaying results.

Doing integration testing developer confirm system components work together. This ensures reliable performance from start to finish and provides smooth user experience during real time operations.

❖ **Acceptance Testing:**

Acceptance testing is done to make sure the system does what users expect its end users. In this project, testing focuses on system identify missing children or criminals by comparing uploaded images, regardless of whether they are recent or old. The testing also checks that the Graphical user interfaces is user friendly allowing anyone to navigates the System without technical skills.

It verifies that the system responds quickly and that the outputs are clear and reliable, providing accurate results in real world situations. Acceptance testing confirms that system fully operational, effective and ready for use. It ensures system meets practical needs of law enforcement agencies social service organizations and other authorized users.

❖ **System Testing:**

System testing checks the complete, integrated application under real world conditions. In this project it looks at the systems ability to detect and recognize faces accurately in different lighting situations, image qualities and facial angles.

This testing evaluate systems performance, reliability, accuracy and response speed ensure meets expected standards during actual use. It also checks cross platform compatibility making sure the application works correctly on various operating systems such as Windows and Linux.

Additionally system testing confirming that the application can manage continuous inputs process images effectively and prevent crashes or errors for better stability and strength. This phase ensures the system is fully ready for deployment and capable of supporting real time face recognition tasks in critical situations.

7.2 Test Case

Test Case ID	Description	Preconditions	Steps	Expected Results	Actual Result	Status
TC01	Verify that the user can upload an Old Image.	Application is running.	1. Click "Old Image" button.2. Select valid image file.3. Upload the image.	Old image should be displayed in the GUI preview panel.	As expected	Passed
TC02	Verify that the user can upload a Recent Image.	Application is running.	1. Click "Recent Image" button.2. Select valid image file.3. Upload the image.	Recent image should be displayed in the GUI preview panel.	As expected	Passed
TC03	Verify that the system detects a face in both images.	Old and recent images uploaded.	1. Click "Analyse" button.2. Wait for processing.	System should detect face in both images, else show "Face Not Detected" message.	As expected	Passed
TC04	Verify that the system matches faces correctly.	Faces detected in both images.	1. Click "Analyse" button.2. View result.	System should display "Face Matched" or "Face Not Matched" with similarity score.	As expected	Passed

TC05	Verify criminal record retrieval for matched faces.	Face matched with criminal database.	1. Run face match.2. Check records.	Criminal details (name, crime, punishment, wanted status) should be displayed.	As expected	Passed
TC06	Verify missing child record retrieval for matched faces.	Face matched with missing child database.	1. Run face match.2. Check records.	Missing child details (name, missing year, guardian contact) should be displayed.	As expected	Passed
TC07	Verify "No Record Found" message for unmatched faces.	Face detected but no database match.	1. Run face match.	System should display "No Records Found".	As expected	Passed
TC08	Verify system handles invalid image formats.	Application is running.	1. Attempt to upload unsupported file format (e.g., .txt).	System should display "Invalid file format" error.	As expected	Passed

TC09	Verify system handles images without faces.	Application is running.	1. Upload image without a face.2. Analyse.	System should display “Face Not Detected”.	As expected	Passed
TC10	Verify exit functionality.	Application is running.	1. Click “Exit” button.	Application should close gracefully.	As expected	Passed

Table 7.2: Test Cases Table

8 CONCLUSION

The Missing Children and Criminal Investigation System uses AI-based face recognition to effectively identify missing individuals and track criminals. By using convolutional networks along feature extraction methods like MTCNN and VGGFace the system achieves high accuracy in recognizing faces even when there is age difference or partial obstructions.

This technology helps law enforcements by reducing the need for manual searches and speeding up investigations. The system has interactive interface allows user upload images easily analyse them, and get real time results making it user friendly. It is built for scalability, quick response and reliable performance across different platforms.

Its ability to process images in various formats and sizes adds flexibility for real world applications. By combining AI powered face recognition with detailed databases of criminals and missing children the system significantly improves both speed and reliability in identification. It aids in proactive crime prevention helps reunite missing individuals with their families and boosts overall public safety.

In short project demonstrate how ai can be useful to law enforcements and humanitarian efforts. With further improvements, larger datasets and integration with government systems the solution could grow into a nationwide platform that enhances security and protects vulnerable individuals.

9 FUTURE ENHANCEMENTS

The system can be improved by integrating better search and matching algorithms. Future improvements for the system could involve using better AI models like deep learning with transformers to improve face recognition in tough situations such as aging, low resolution images, partial blockages or disguises. Adding 3D face reconstruction could also enhance matching accuracy over time by capturing facial features in greater detail.

Another possible improvement is linking the system to national and international databases. Connecting with centralized police records, Interpol databases and missing persons lists would expand search capabilities, assisting in tracking criminals and finding missing children across different regions and countries.

The system gain real time inputs from CCTV cameras and drones making it more effective in practical situations. By using real time streams from surveillance camera and drones AI models can continually scan public areas and automatically detect faces, which will improve monitoring efficiency and public safety.

Additionally, a mobile app for field officers can enable them to capture and compare images instantly on site. This mobile capability would speed up investigations, improve response times and enhance operational efficiency.

Finally integrating predictive analytics for criminal activity can offer authorities insights into possible crime patterns. By looking at historical and real time data the system can help with preventive measures and planning the strategic use of resources.

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